Prática de Física dos Dispositivos Eletrônicos

FGA0100

Laboratório-8

Efeitos Peltier e Seebeck

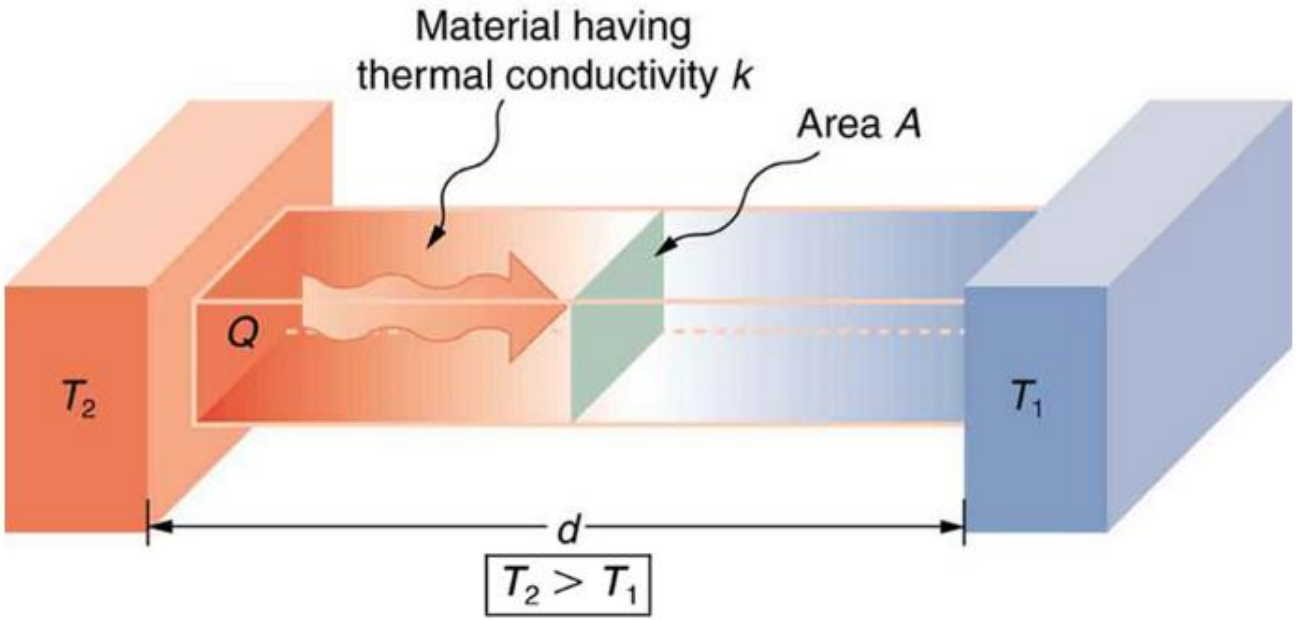
01/2022

FGA **Universidade de Brasília**

Lei de Fourier (1822)

Propagação de Calor e Condutividade Térmica

Jean-Baptiste Joseph Fourier
Físico Matemático Francês
(1768 - 1830)



$$\vec{q} = -k\nabla T$$

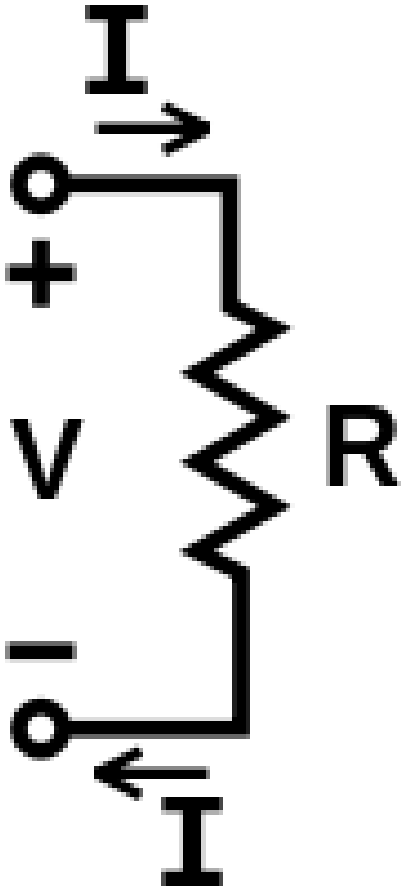
$\vec{q}$ is the local heat flux density, $W \cdot m^{-2}$

k is the material's conductivity, $W \cdot m^{-1} \cdot K^{-1}$,

∇T is the temperature gradient, $K \cdot m^{-1}$.

Lei de Ohm (1827)

Georg Simon Ohm
Físico Alemão
(1789–1854)



$$I = \frac{V}{R}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

$$R = \rho \frac{l}{a}$$

Termoeletricidade

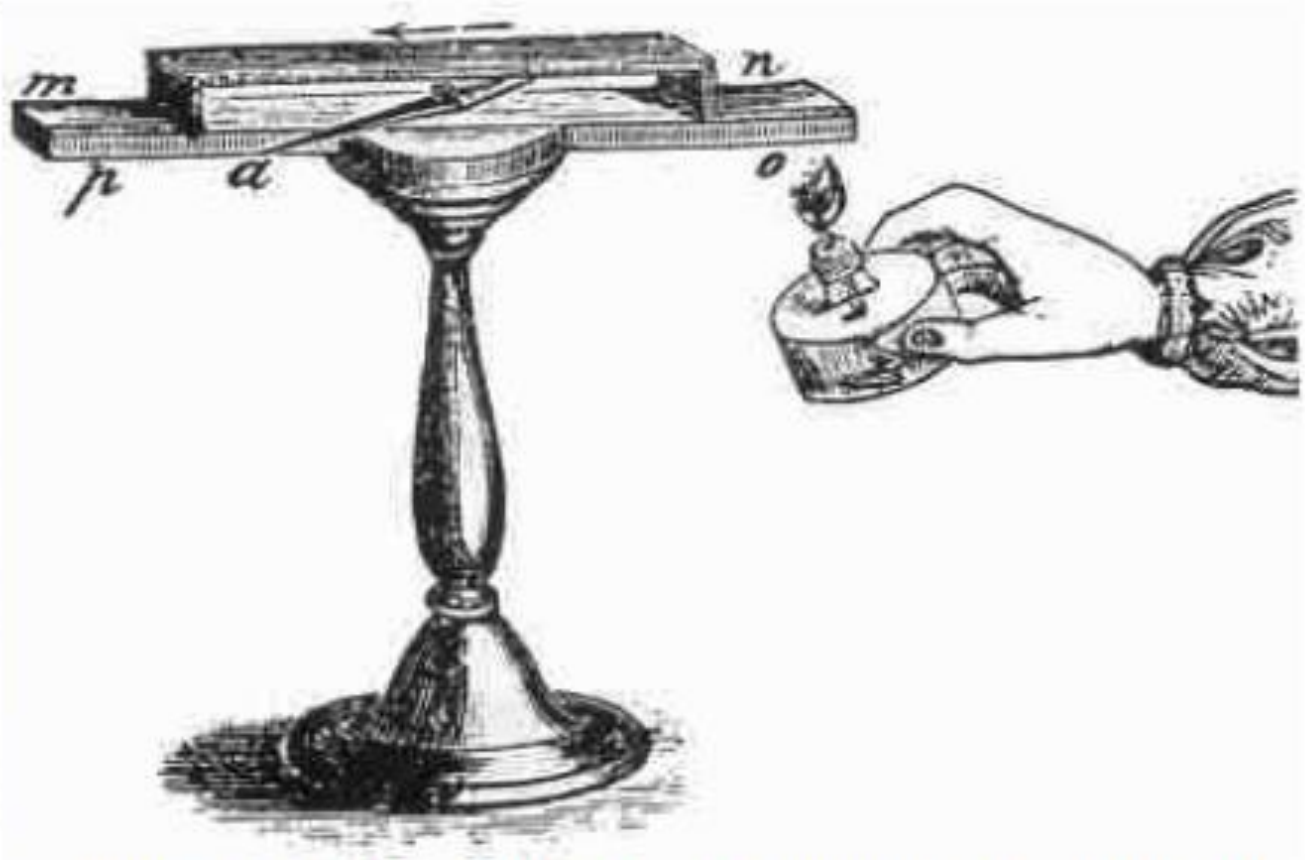
Georg Simon Ohm
Físico Alemão
(1789–1854)

Termopar de Cobre e Bismuto

Experimento: Efeito Seebeck (1821)

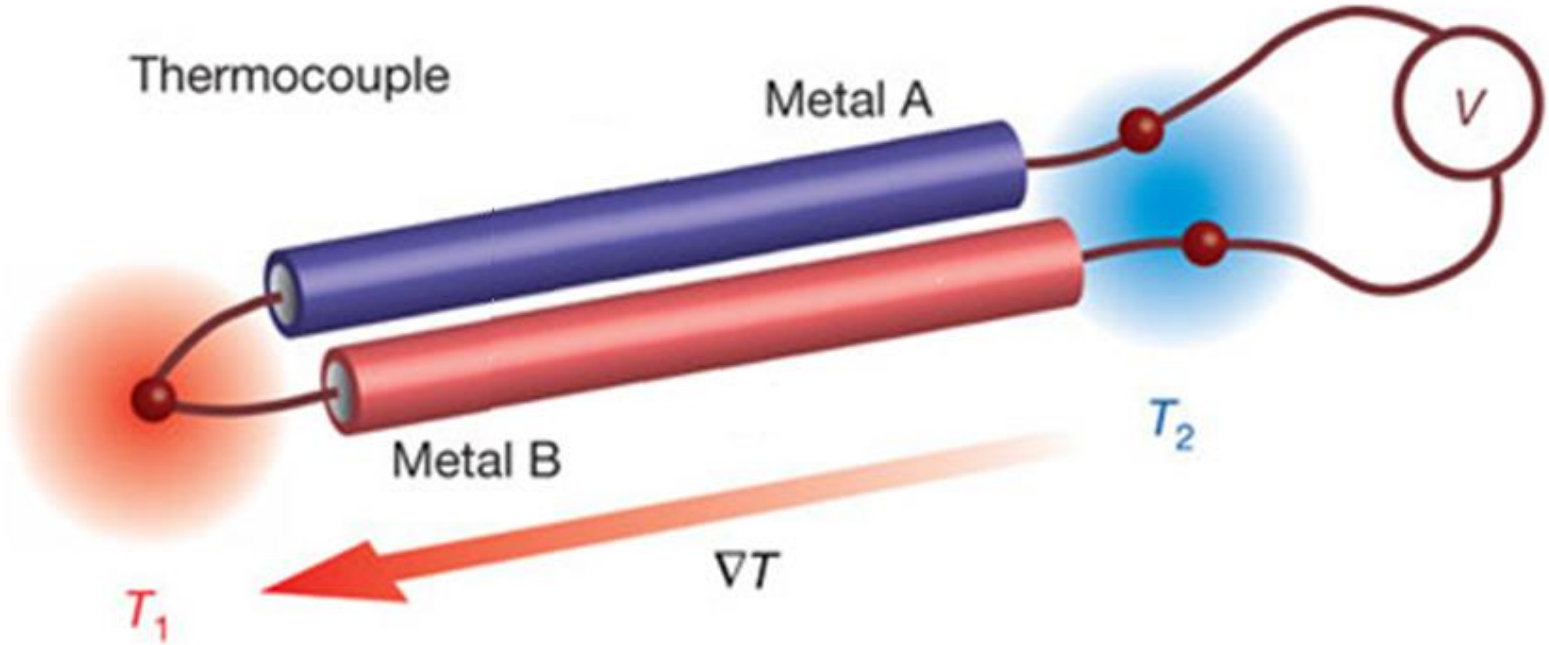
Thomas Seebeck

Físico Alemão
(1770-1831)



Instrument used by Seebeck to observe the deflection of a compass needle (a) due to a thermoelectric induced current from heating the junction of two different metals (n and o).

Efeito Seebeck (1821)



$$\vec{J} = -\sigma \nabla V - \sigma S(T) \nabla T$$

$$\Downarrow \quad \vec{J} = 0$$

$$\nabla V = -S(T) \nabla T$$

Efeito Seebeck
(Em Equilíbrio Térmico)

$$\nabla V = -S(T)\nabla T \qquad \frac{\partial S}{\partial T} \approx 0$$

$$\begin{cases} \textbf{Metal-X:} \quad \dfrac{\Delta V_X}{\Delta x} \approx -S_X \dfrac{\Delta T_X}{\Delta x} \\[2ex] \textbf{Metal-Y:} \quad \dfrac{\Delta V_Y}{\Delta x} \approx -S_Y \dfrac{\Delta T_Y}{\Delta x} \end{cases}$$

Equilíbrio: $\Delta T_X = \Delta T_Y = T_h - T_c$

Efeito Seebeck
(Em Equilíbrio Térmico)

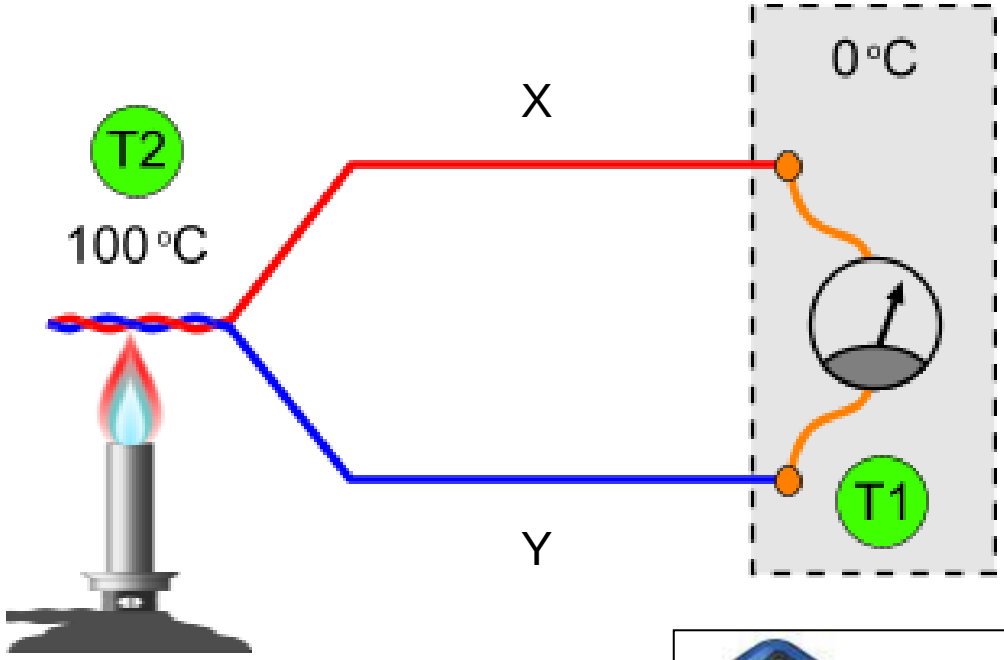
$$\begin{cases} (V_0 - V_Y) = \Delta V_Y = -S_Y (T_h - T_c) \\ (V_0 - V_X) = \Delta V_X = -S_X (T_h - T_c) \end{cases}$$

Subtrair

$$(V_X - V_Y) = (S_X - S_Y)(T_h - T_c)$$

$$\Delta V = V_X - V_Y = K_{XY}(T_h - T_c)$$

Termopar com Multímetro



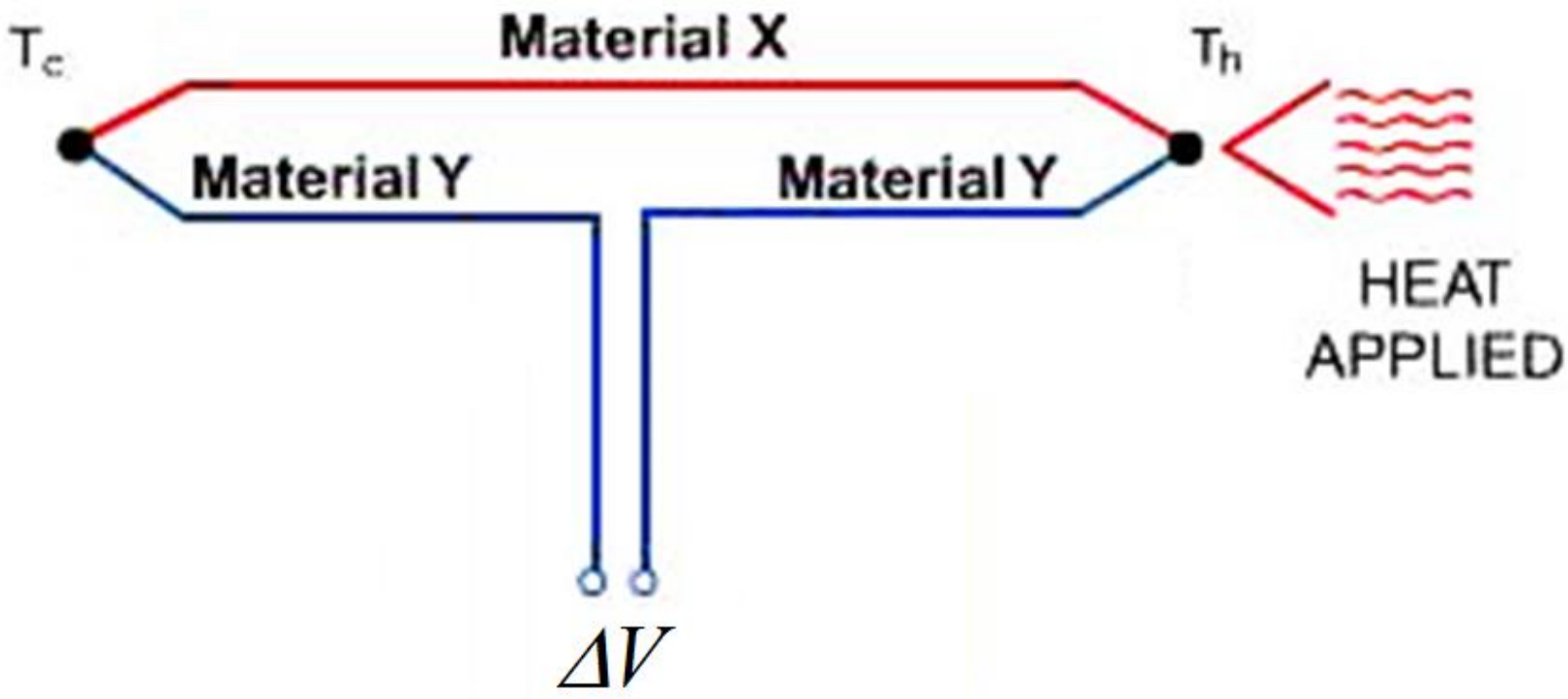
Medição Interna Multímetro

$$\Delta V = K_{XY}(T_2 - T_1)$$

http://www.pcbheaven.com/wikipages/How_Thermocouples_Work/

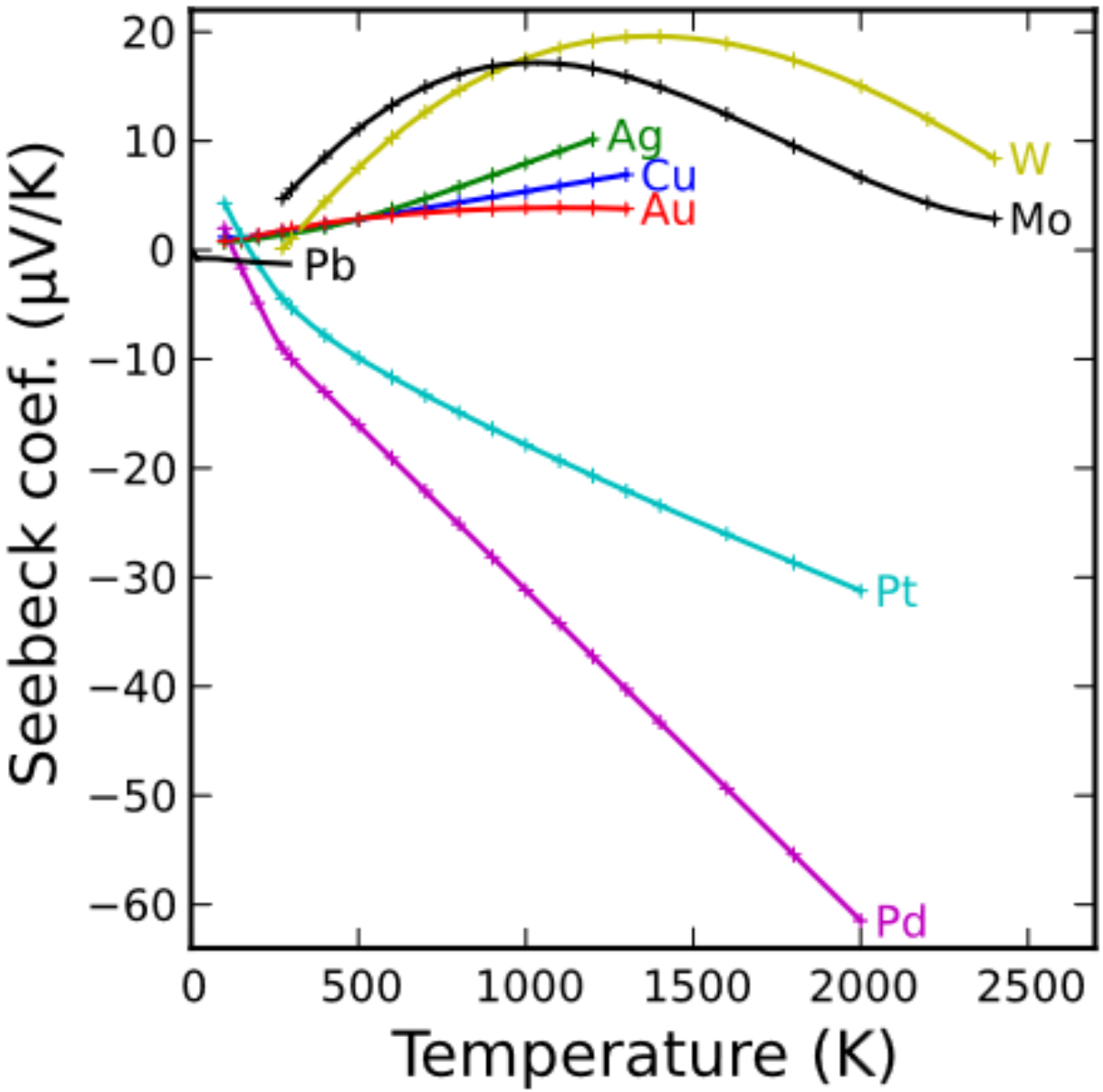
Efeito Seebeck

Termopar



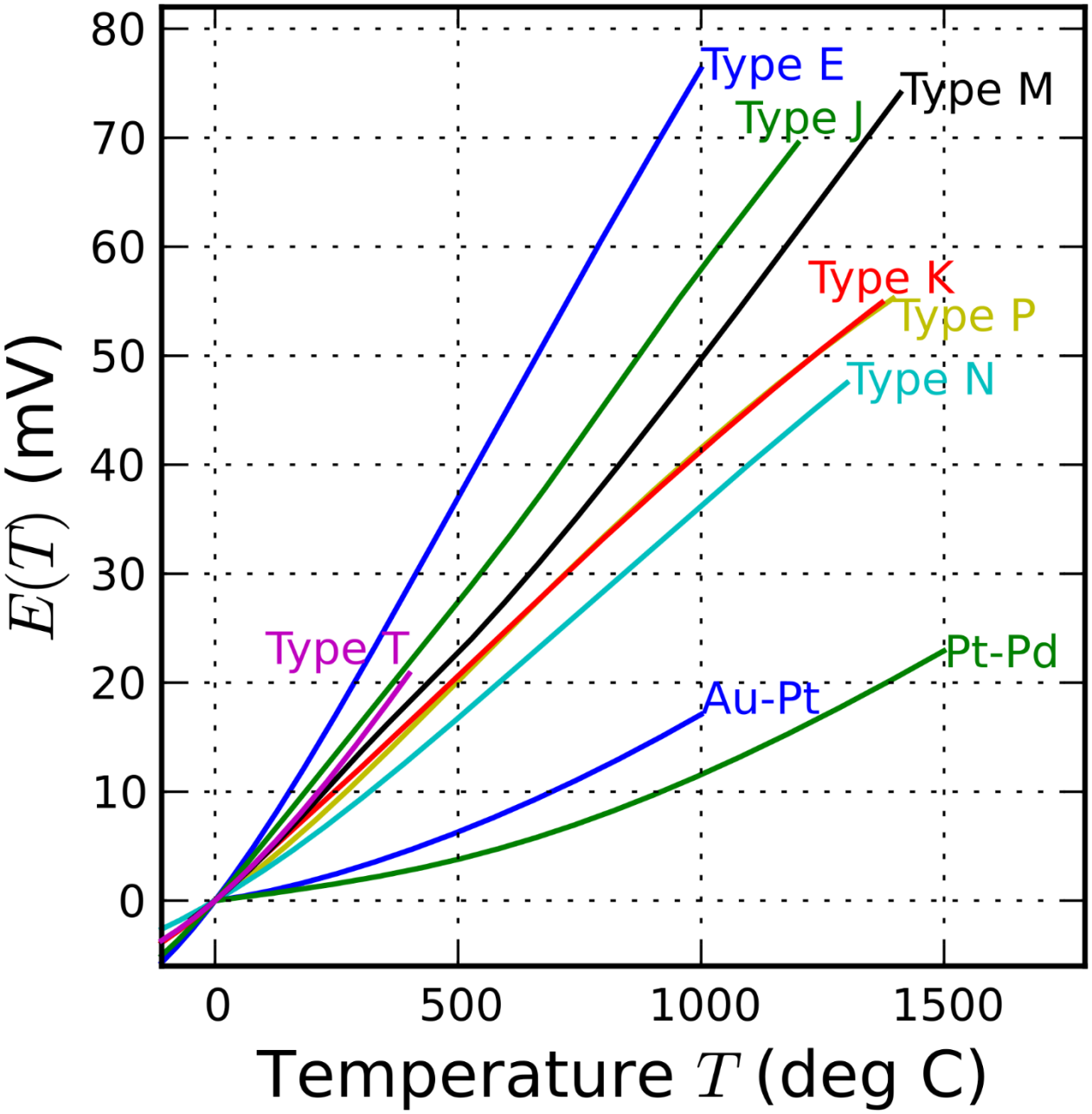
$$\Delta V = K_{XY}(T_h - T_c)$$

Hot **Cold**

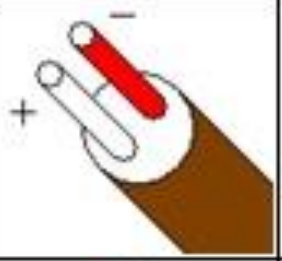
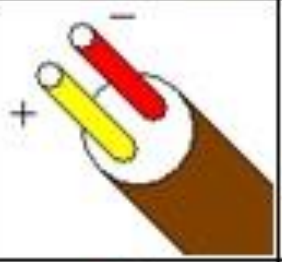
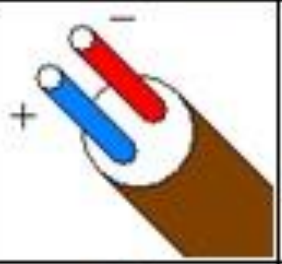
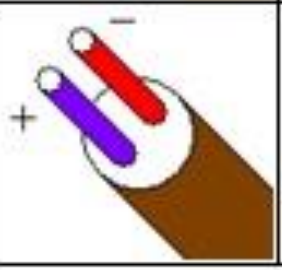


Absolute Seebeck coefficients of various metals up to high temperatures, mainly from Cusack & Kendall (1958). The data for lead (Pb) is from Christian, Jan, Pearson, Templeton (1958).

Tipos de Termopares

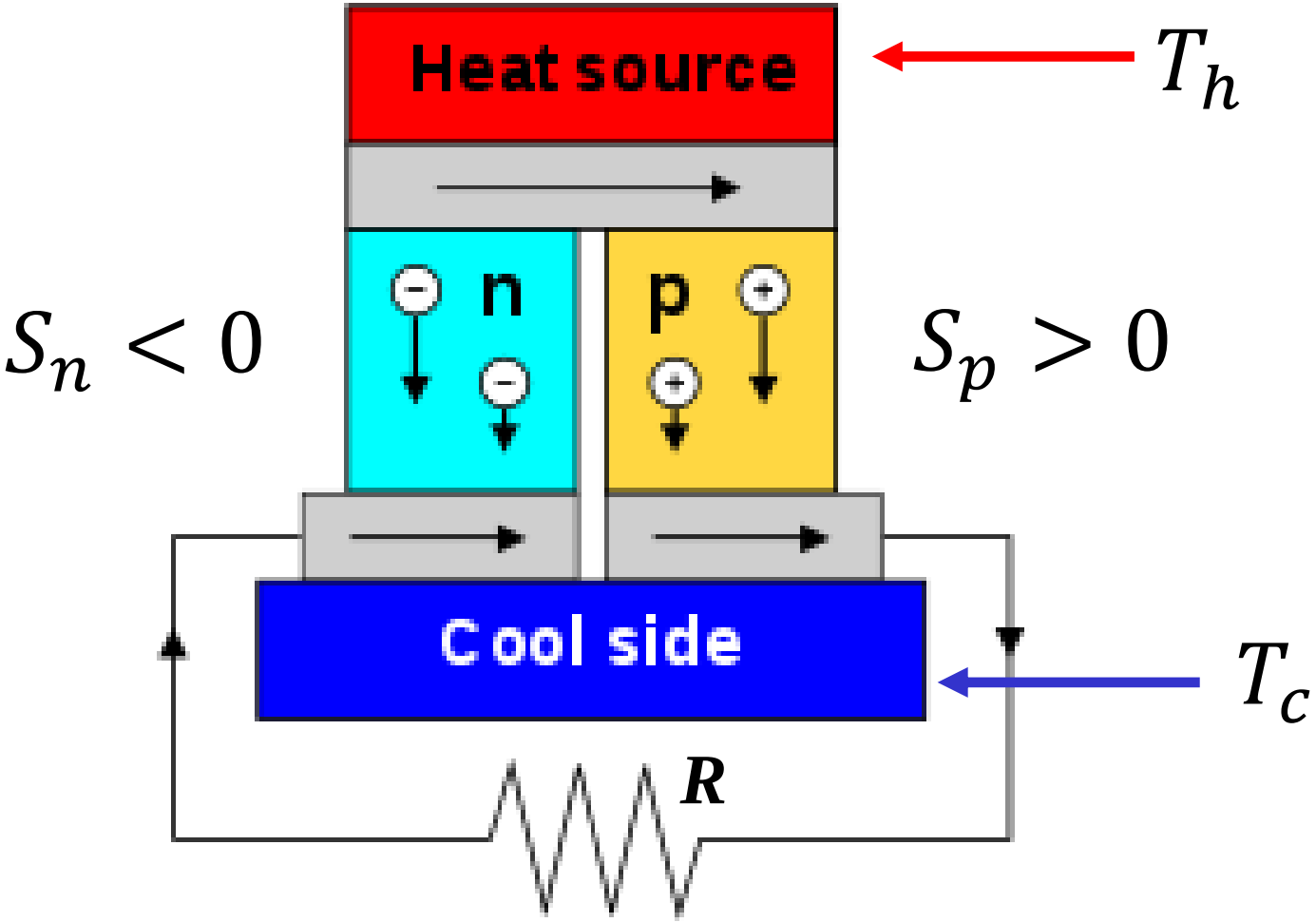


Tipos de Termopares

Type	Material		Color Code	Range (°C)	
Thermocouple Grade	Positive Wire	Negative Wire		Minimum	Maximum
J	Iron	Constantan		0	750
K	Chromel	Alumel		-200	1250
T	Copper	Constantan		-200	350
E	Chromel	Constantan		-200	900

Efeito Seebeck

Semicondutores



$$\Delta V = RI = (S_p - S_n)(T_h - T_c) = S(T_h - T_c)$$

Assumindo que a resistência interna da Célula Peltier é muito pequena comparada com a resistência de carga R.

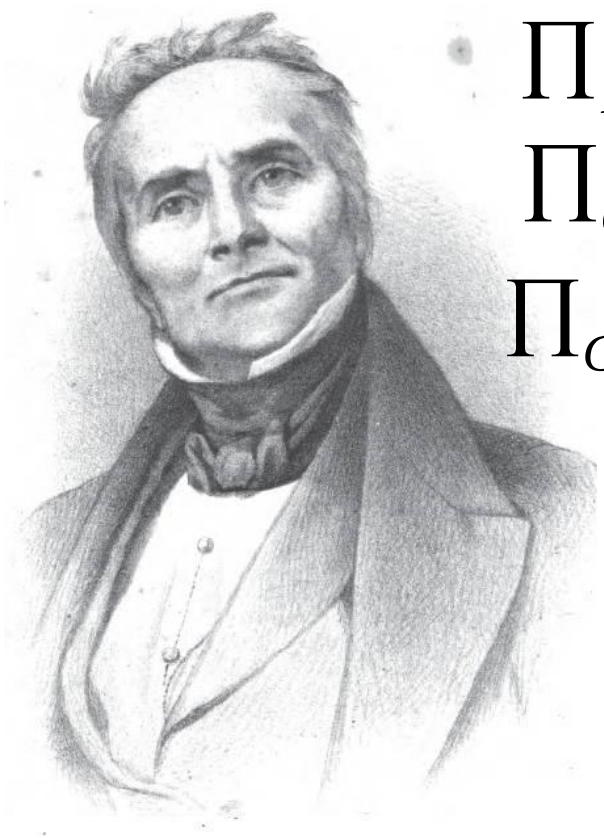
Experimento: Efeito Peltier (1834)

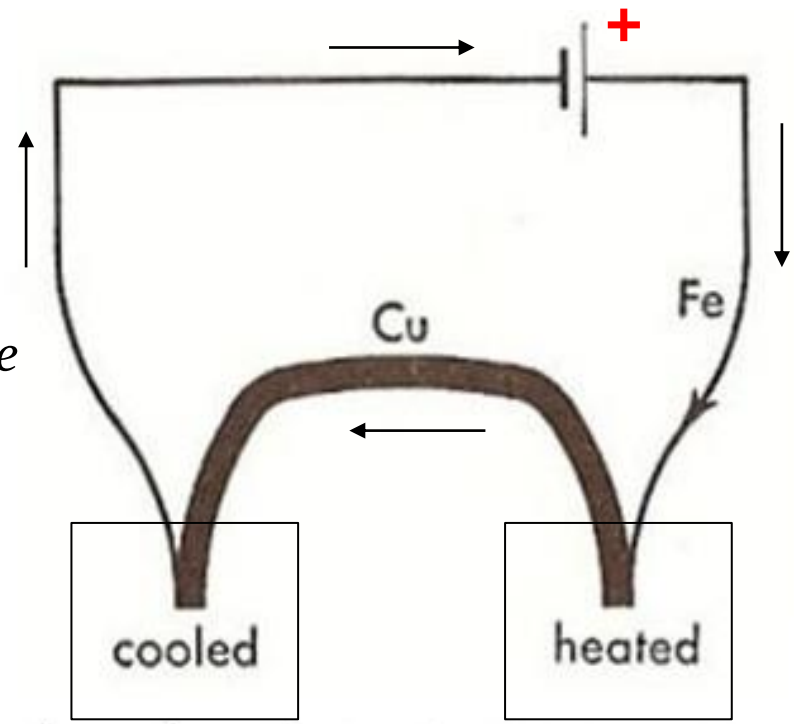
Metais:

$$\Pi_{Fe} > 0$$

$$\Pi_{Cu} > 0$$

$$\Pi_{Cu} < \Pi_{Fe}$$





From Duckworth, *Electricity & Magnetism*, p 181

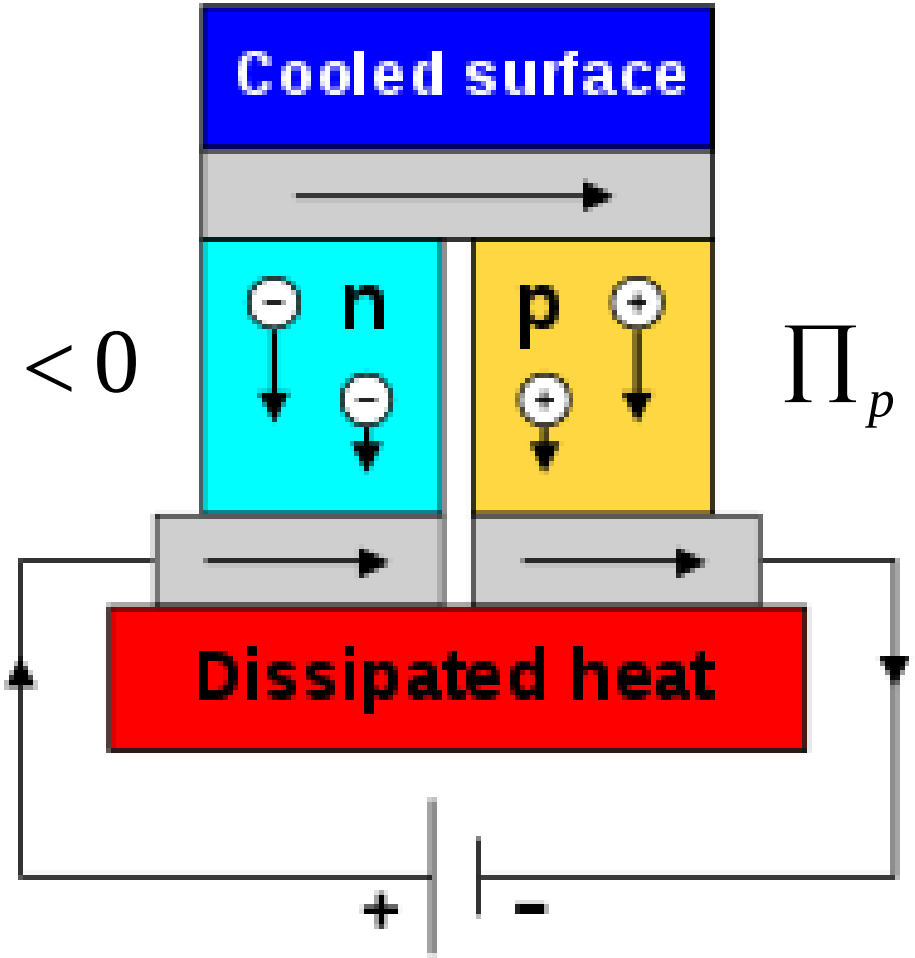
Jean Charles Athanase Peltier
Físico Francês
(1785 – 1845)

$$\frac{\partial Q}{\partial t} = (\Pi_{Fe} - \Pi_{Cu})I = \Pi I$$

Efeito Peltier

Semicondutores

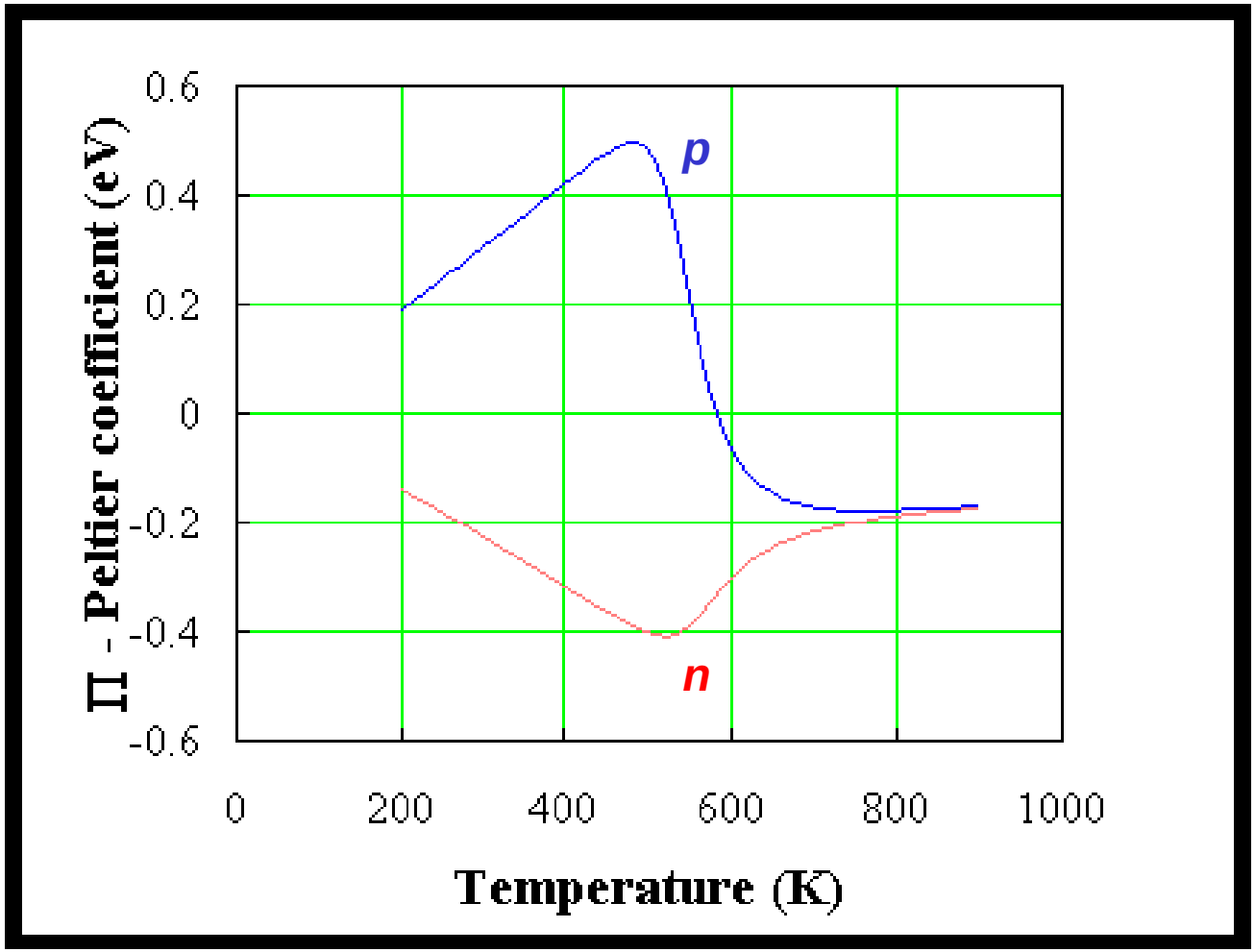
$\Pi_n < 0$



$\Pi_p > 0$

$$\frac{\partial Q}{\partial t} = (\Pi_p - \Pi_n) I = \Pi I$$

Coeficientes Peltier vs. Temperatura
Silício



Dopado: 10^{14} [cm^{-3}]

https://ecee.colorado.edu/~bart/book/tepower.htm

Efeito Thomson (Lord Kelvin - 1854)

Densidade de Corrente

Taxa de Geração ou Absorção de Calor: $$\dot{q} = -\mathcal{K}\mathbf{J}\cdot\nabla T$$

Coeficiente de Peltier

Coeficiente de Thomson

Gradiente de Temperatura

$$\begin{cases} \mathcal{K} \equiv \dfrac{d\Pi}{dT} - S \\ \Pi = TS \end{cases}$$

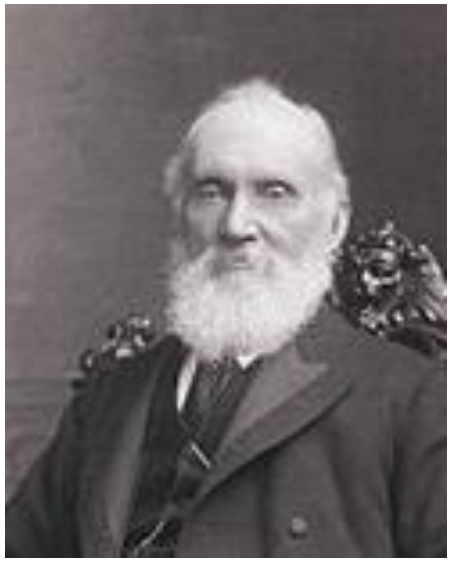
Primeira Relação de Thomson

Segunda Relação de Thomson

Coeficiente de Seebeck

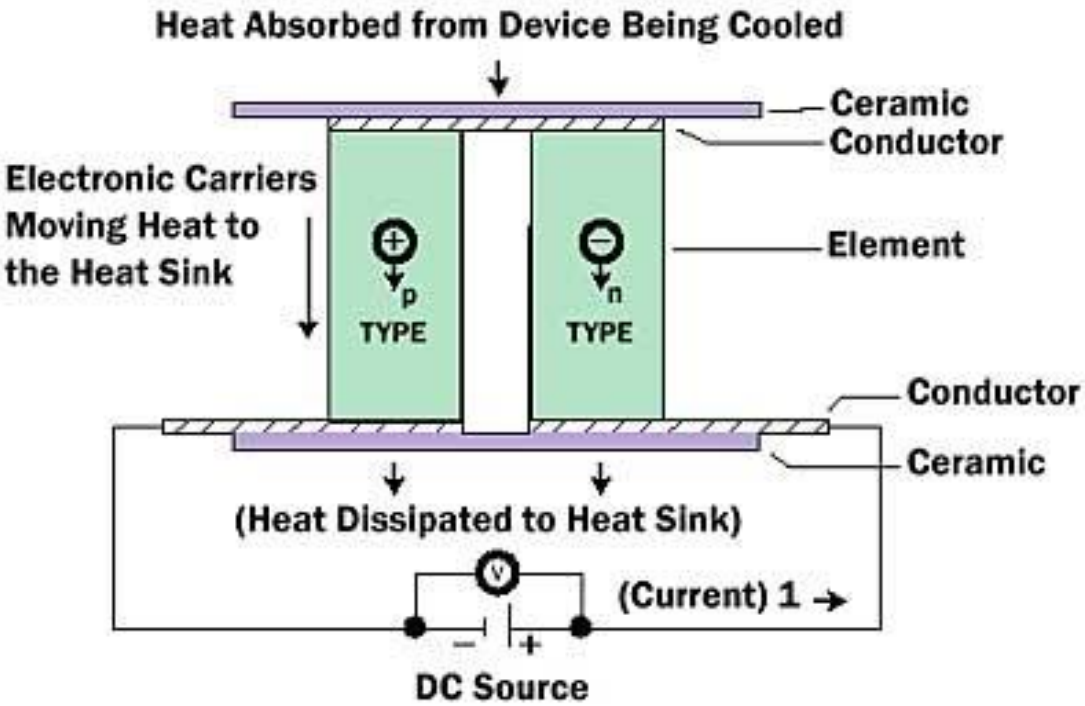
$$\boxed{\mathcal{K} = T\frac{dS}{dT}}$$

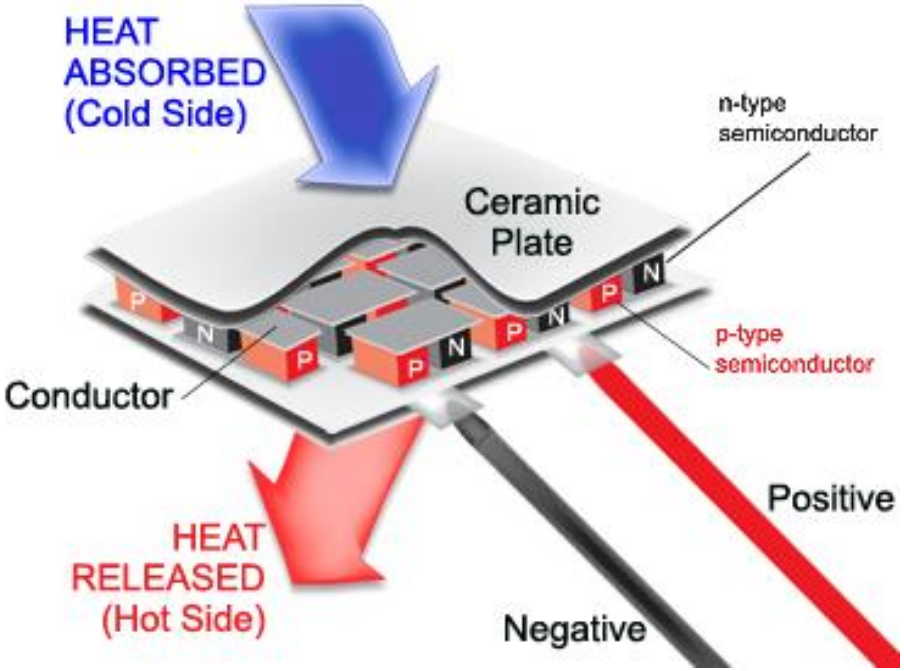
Não incluindo a condução térmica e aquecimento por efeito Joule.

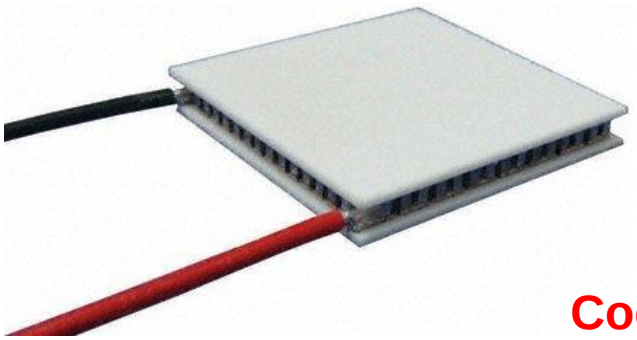


William Thomson
Lord Kelvin
Físico, Matemático e Engenheiro Britânico
(1824 – 1907)

Dissipação de Calor por Módulo Peltier







Poder de Resfriamento (*Cooling Power*):

$$\frac{\partial Q_c}{\partial t} = ST_c I - \frac{1}{2} I^2 R - K(T_h - T_c)$$

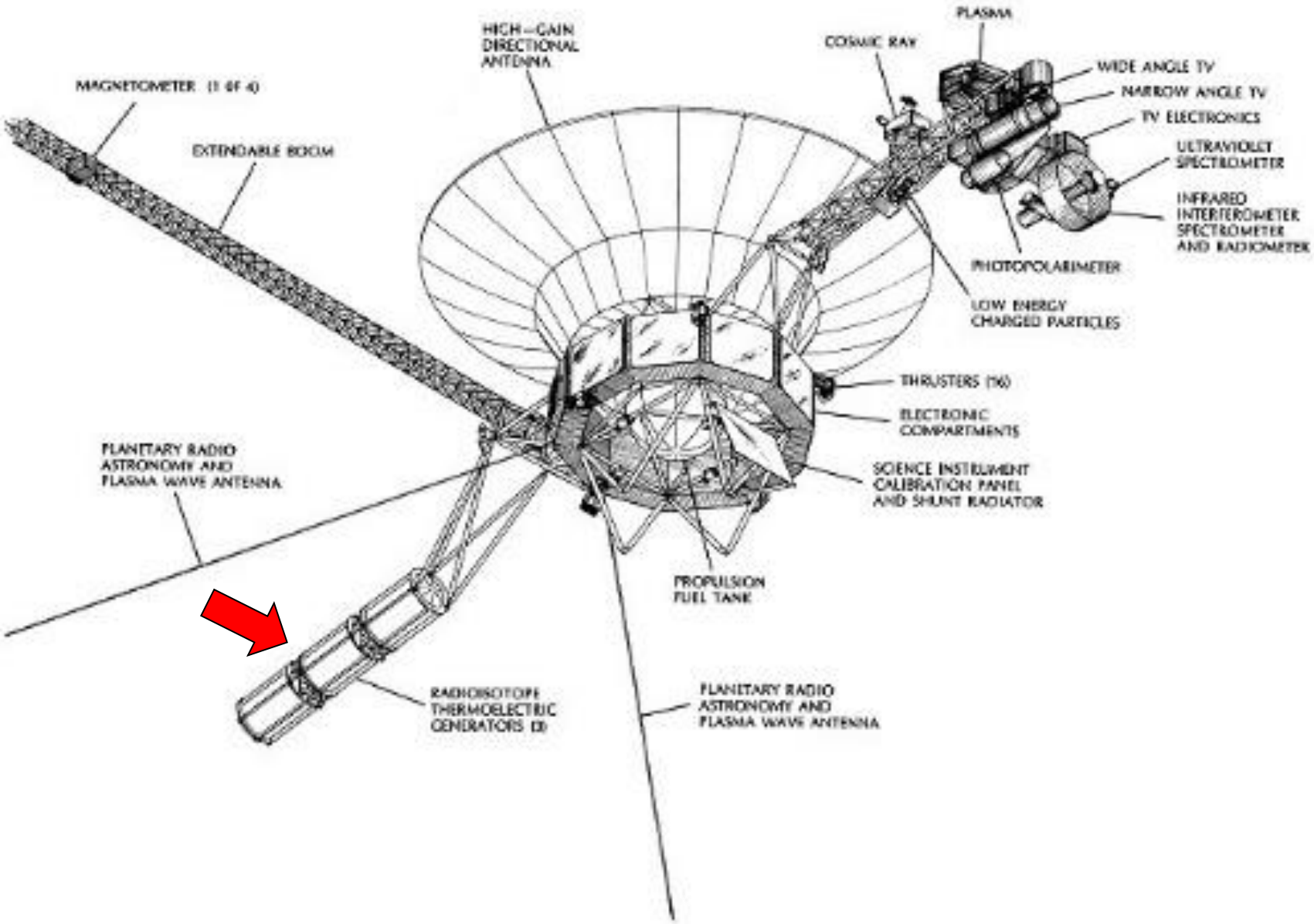
Coeficiente Seebeck

Efeito Joule

Condução Térmica

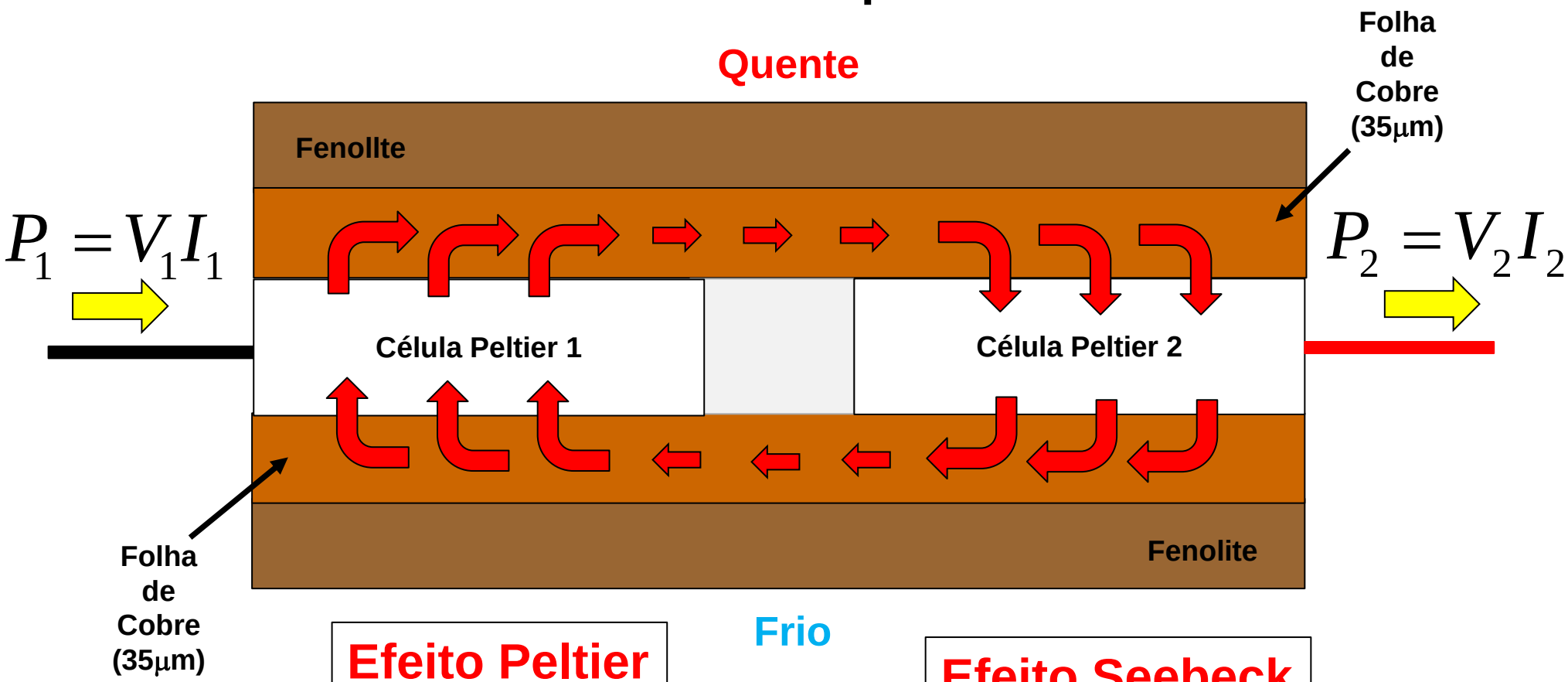
Coeficiente Peltier: (junção fria) $\Pi_c = ST_c$

Gerador Termoelétrico (Radioisótopos)
Sondas Espaciais Voyager I e II



Arranjo Experimental

Par de Módulos Peltier Acoplados Termicamente



Efeito Peltier

Efeito Seebeck

Eficiência de Conversão: $P_2 = \eta_P \eta_T \eta_S P_1$

Eficiência Térmica

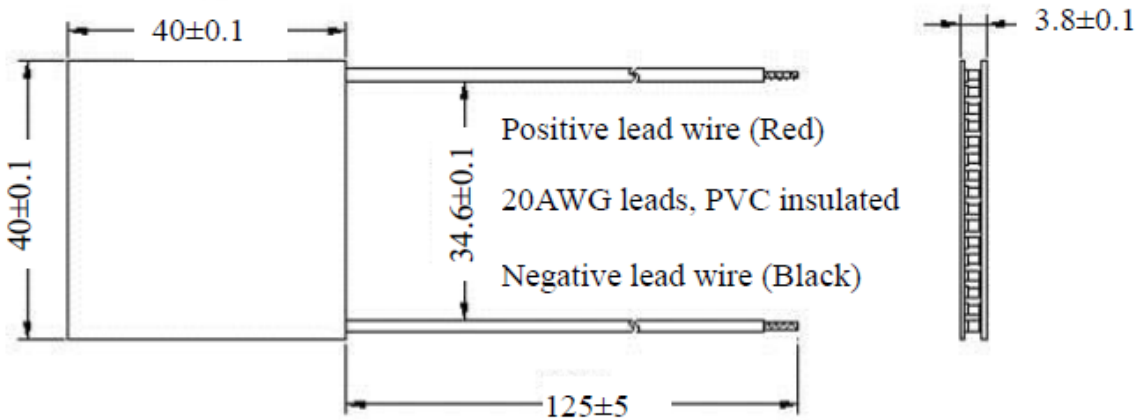
$$P_2 = \eta_{eff} P_1$$

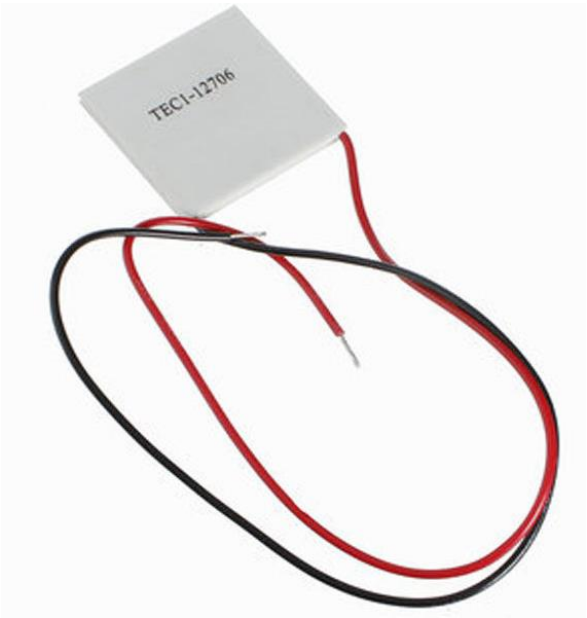
Módulo Peltier (TEC1-12706)

Performance Specification Sheet

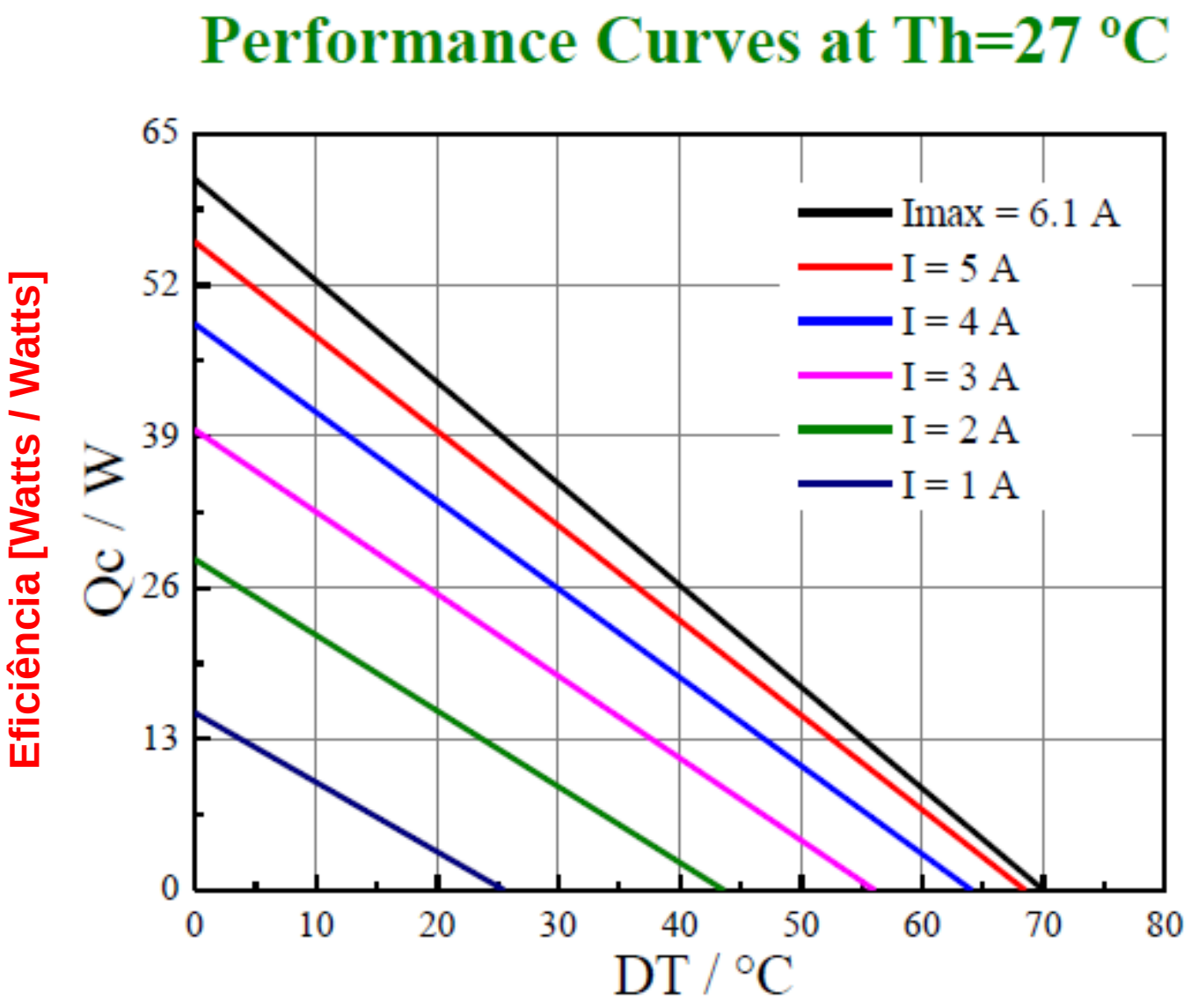
Th(°C)	27	50	Hot side temperature at environment: dry air, N_2
DT_{max}(°C)	70	79	Temperature Difference between cold and hot side of the module when cooling capacity is zero at cold side
U_{max}(Voltage)	16.0	17.2	Voltage applied to the module at DT_{max}
I_{max}(amps)	6.1	6.1	DC current through the modules at DT_{max}
Q_{Cmax}(Watts)	61.4	66.7	Cooling capacity at cold side of the module under DT=0 °C
AC resistance(ohms)	1.8 ~ 2.2	2.0~2.4	The module resistance is tested under AC

Geometric Characteristics Dimensions in millimeters





Módulo Peltier (TEC1-12706)

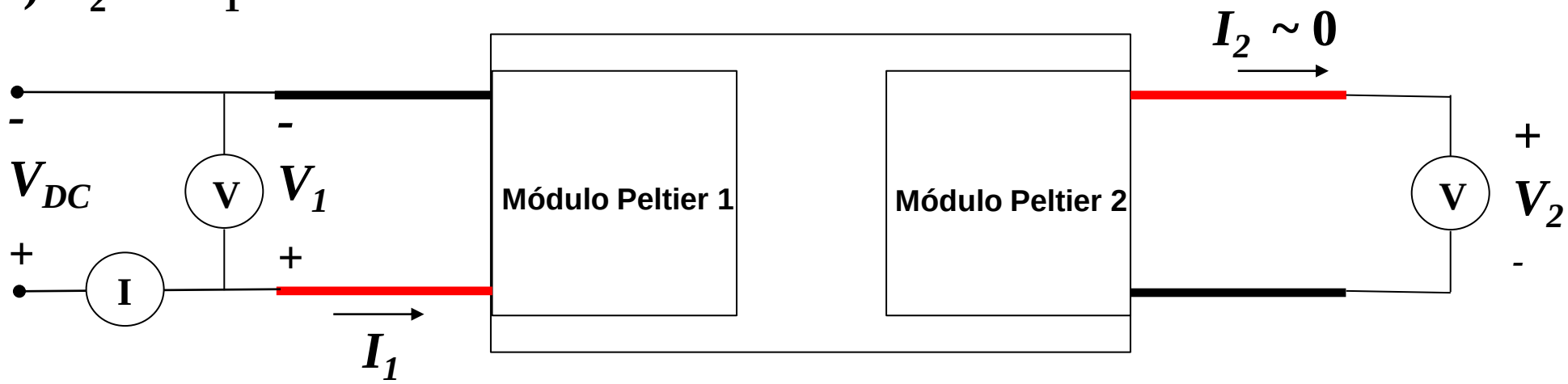


Eficiência [Watts / Watts]

Diferença de Temperatura

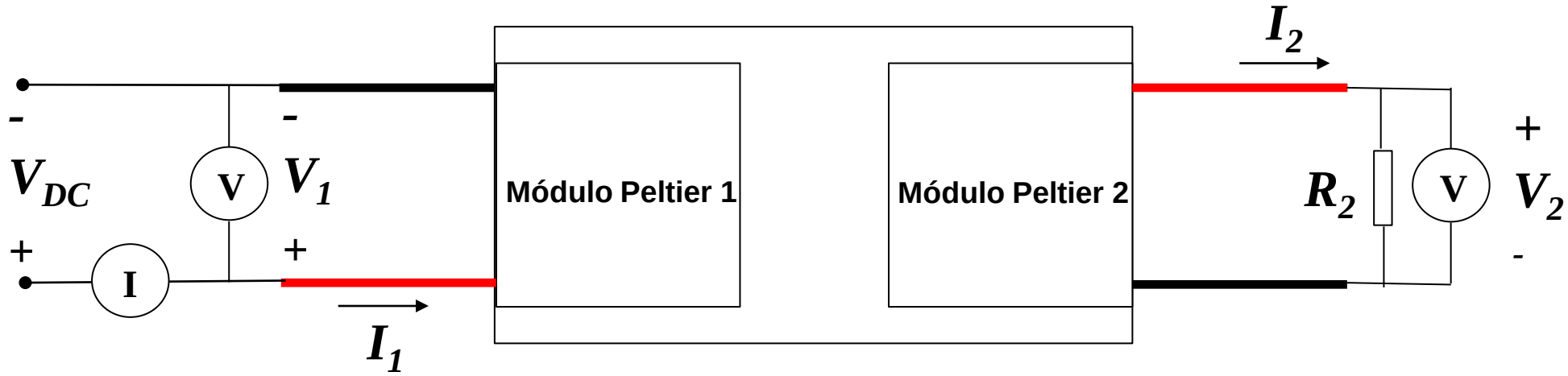
Procedimentos Experimentais

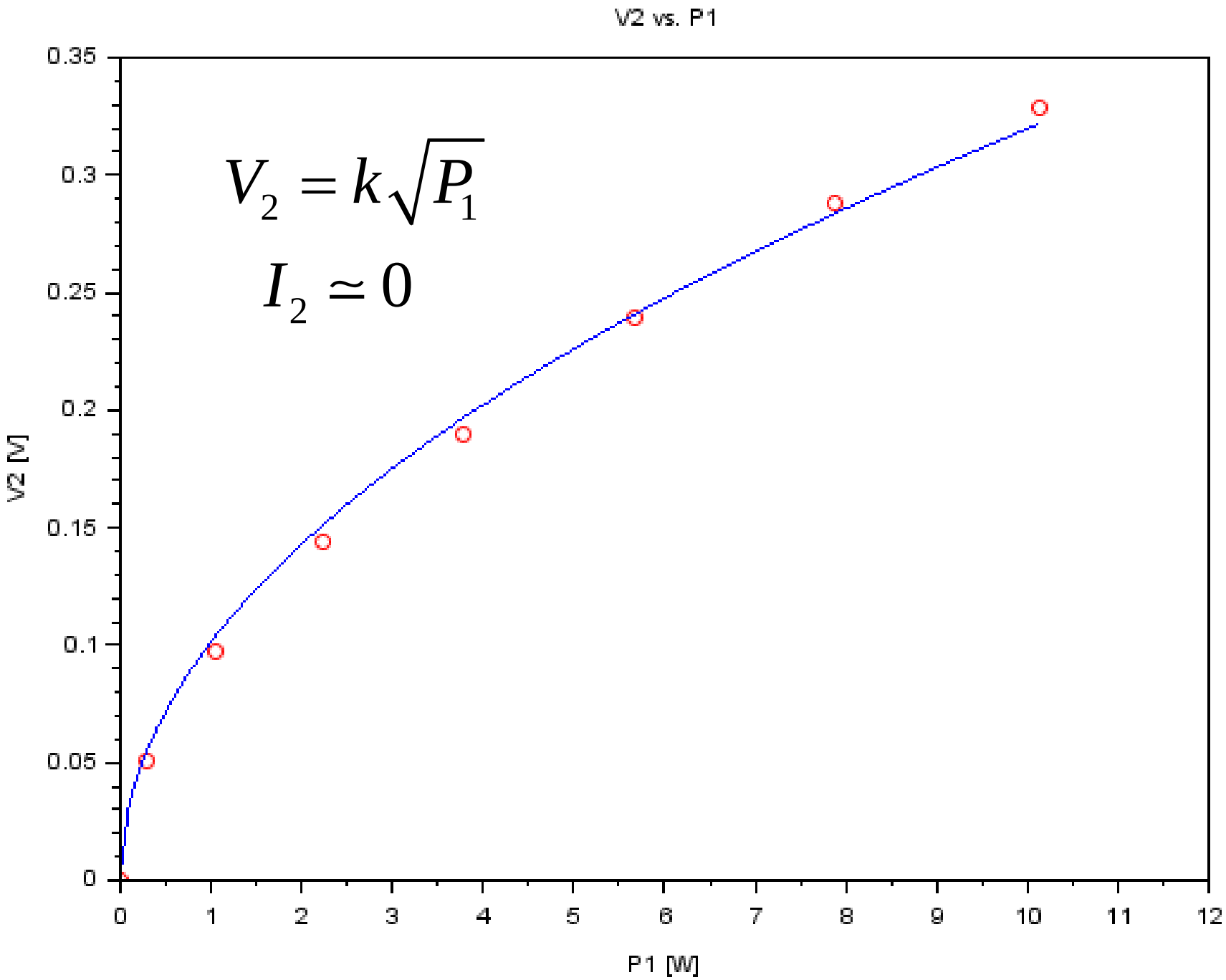
I) V_2 vs. P_1

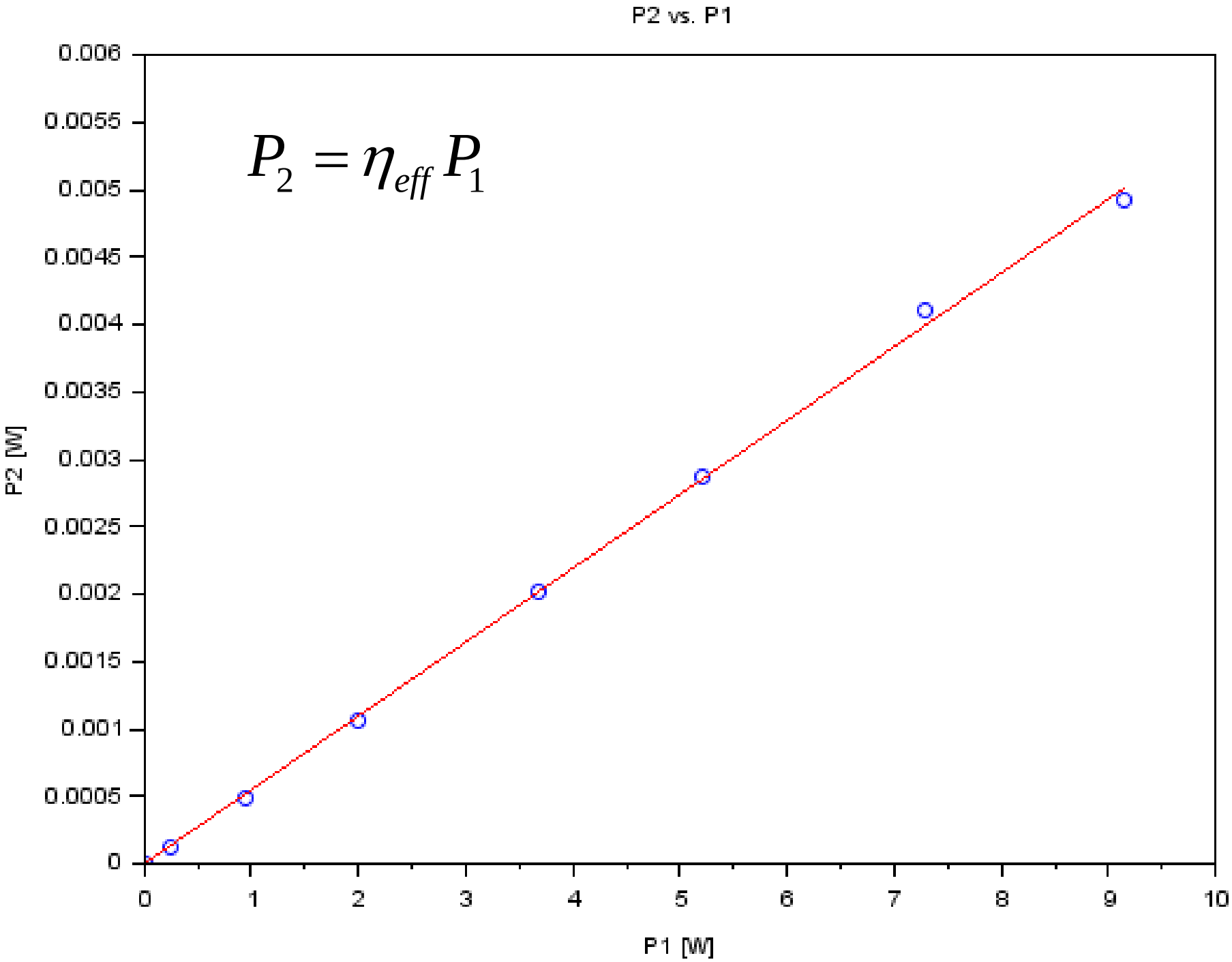


$$P_1 = V_1 I_1 \qquad P_2 = V_2 I_2$$

II) P_2 vs. P_1




V2 vs. P1
V2 [V]
P1 [W]
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
1
2
3
4
5
6
7
8
9
10
11
12
$V_2 = k\sqrt{P_1}$
$I_2 \simeq 0$


P2 vs. P1
0.006
0.0055
0.005
0.0045
0.004
0.0035
0.003
0.0025
0.002
0.0015
0.001
0.0005
0
P2 [W]
0
1
2
3
4
5
6
7
8
9
10
P1 [W]

$$P_2 = \eta_{eff} P_1$$